National University of Lesotho

Department of Mathematics and Computer Science

CS4433 Software Engineering

Individual Final Project

Moderated Learning Management System

Deployment

Version 1.0

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1. Introduction

This document is the deliverable of Phase 7. It has everything that is needed to put version 1.0 of the Moderated LMS on a school computer and to use it.

Table 1.1: The deliverables of Phase 7

Deliverable	Where it is	Remark
UML deployment diagram	Section 2	
Installation guide for a school technician	Section 3	Written for NFR-11 (install and start in 30 minutes)
User manual for the Student	Section 4.2	Written for NFR-07 (people who are not IT specialists)
User manual for the Lecturer	Section 4.3	
User manual for the Head of Department	Section 4.4	
User manual for the Administrator	Section 4.5	
Release notes	Section 5	The same text is in docs/RELEASE_NOTES.md
Tag v1.0	Section 6	In the Git repository

2. Deployment diagram

Figure 2.1 shows where the parts of the system are placed. The system is on one school computer, as in assumption A2 of the SRS. It works without internet (NFR-05).

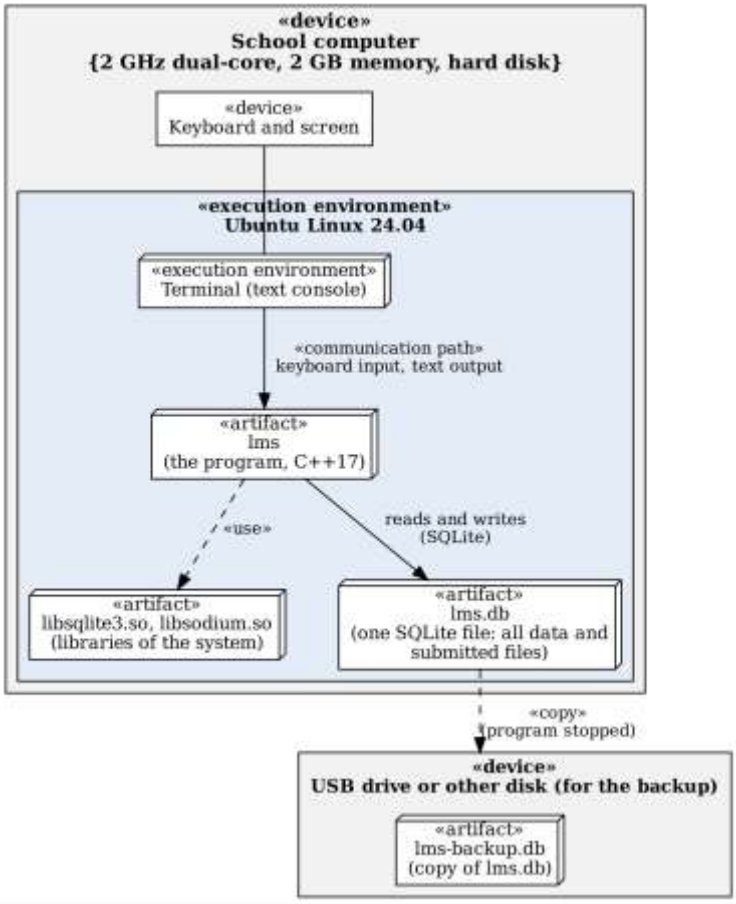


Figure 2.1: Deployment diagram of the Moderated LMS, version 1.0

Table 2.1: The parts of the diagram

Part	What it is
School computer	A normal personal computer with a keyboard, a screen and a hard disk. The speed targets are for a 2 GHz dual-core processor with 2 GB of memory (assumption A6). One person uses the system at a time.
Ubuntu Linux 24.04	The operating system. The program was built and tested on it.
Terminal (text console)	The window where the program is typed into and where its text appears. There is no graphical interface (IF-01).
lms	The program, one file. It holds all four layers. The database schema is built into it.
libsqlite3 and libsodium	Two libraries of the operating system. SQLite stores the data, and libsodium makes the password hashes.
lms.db	The one SQLite file with all data: users, courses, assignments, marks, the audit log and the submitted files.
Backup disk and lms-backup.db	A copy of lms.db on a USB drive or another disk. It is made while the program is stopped (Section 3.5).

There is no network connection in the diagram because the system does not use one. The internet is needed only once, to install the tools of Section 3.2.

3. Installation guide

This guide is for a school technician. Follow the steps in order. The steps were tested on a clean copy of the repository: the build took about 30 seconds. A timed trial with a technician (NFR-11) has not been done yet; the limit is 30 minutes.

3.1 What you need

- A computer with **Ubuntu Linux 24.04** (the version that was tested; 22.04 should also work but was not tested), a keyboard, a screen and at least 200 MB of free disk space.
- A user account that can use **sudo** (the password of an administrator of the computer).
- The internet, **only for Step 2 and Step 3**. After that the system works with the cable out.
- The address of the Git repository (Section 2.1 of the Phase 6 document), or the file lms_source.zip.

3.2 Steps

Step 1. Open the terminal. Press Ctrl + Alt + T.

Step 2. Install the tools. Type this and press Enter. Type the password when it is asked for (nothing shows while you type). It takes a few minutes.

```
sudo apt update
sudo apt install -y git build-essential cmake pkg-config libsqlite3-dev
libsodium-dev
```

Step 3. Get the code. Use one of the two ways.

```
git clone <the address of the repository> lms

or, with the zip file:
unzip lms_source.zip
```

Step 4. Build the program.

```
cd lms
cmake -S . -B build -DLMS_BUILD_TESTS=OFF
cmake --build build
```

The last line on the screen must be **[100%] Built target lms**. If there is a line with the word error, see Section 3.6.

Step 5. Install the program for all users.

```
sudo cp build/lms /usr/local/bin/lms
```

Step 6. Make the folder for the data.

```
mkdir ~/lms-data
```

Step 7. Start the program for the first time.

```
lms --db ~/lms-data/lms.db
```

The program makes the database file and asks for the first Administrator: type the username, the full name and the password twice. The password needs at least 8 characters, with a letter and a digit. **There is no default password.** Write the password in a safe place.

Step 8. Check that it works. At "Choose:" type 1 and Enter, log in with the new account, check that the Administrator menu appears, then type 0 to log out and 0 again to exit. The text "Goodbye." shows.

Step 9. Make a second Administrator (menu 2, role 1) and write its password too. If the only Administrator forgets the password, nobody can reset it.

The system is installed. To start it later, open the terminal and type the command of Step 7.

3.3 A practice copy (only for training)

To show the system with demo users, use a different file and never the real one:

```
lms --db /tmp/practice.db --demo
```

The demo users are admin (password Admin1234) and lecturer1, hod_cs, student1 and others (password Demo1234). Never use --demo for real data.

3.4 Update to a new version

Stop the program, make a backup (Section 3.5), get the new code, repeat Steps 4 and 5. The data file stays as it is. A data file of a different version number is refused by the program, so nothing is damaged.

3.5 Backup and restore

All data, including the submitted files, is in the one file lms.db. Make a copy every day when the program is not running:

```
cp ~/lms-data/lms.db /media/<your-usb-drive>/lms-backup-2026-11-30.db
```

To restore, stop the program and copy the backup over lms.db. Keep at least the last 7 copies.

3.6 If something goes wrong

Table 3.1: Problems and what to do

What you see	What to do
cmake: command not found	Step 2 was not done or did not finish. Do Step 2 again.
Could NOT find SQLite3, or Package 'libsodium' not found	Install the libraries: sudo apt install -y libsqlite3-dev libsodium-dev pkg-config. Then repeat Step 4.
lms: command not found	Step 5 was not done. Run it again from the lms folder.
"The program cannot start: Cannot open the database file ..."	The folder in --db does not exist. Make it (Step 6) or check the spelling.
"This file is a database, but not one of this program. Nothing was changed."	The --db path points to the wrong file. Use the right path. Do not repair the file.
"... unknown version"	The data file is from another version of the program. Use the matching program version or a backup.
A user is locked	After 5 wrong passwords the account is locked for 5 minutes. Wait, then try again. An Administrator can also set a new password.

4. User manuals

There is one manual for each role. Section 4.1 is for everybody. The manuals are short and use the same words as the menus of the program.

4.1 Before you start (everyone)

- The system is used with the keyboard only. Type the number of a choice and press **Enter**.
- Start: the technician tells you the command. At "Choose:" type **1**, then your username and password. Nothing shows while you type the password.
- After 5 wrong passwords the account is locked for 5 minutes.
- Dates are typed like this: **2026-11-30 17:00** (year-month-day, a space, hour:minute).
- If you type something wrong, the program says why and asks again. After 3 wrong tries it cancels the action and goes back to the menu. Nothing is saved when an action is cancelled.
- A question "(y/n)" is answered with y for yes or n for no.
- When you log in, messages for you (for example that a mark was released) are shown first.
- Type **0** to log out, and **0** at the first menu to exit.

4.2 Student

Menu: 1 My courses and assignments, 2 Submit (or resubmit) an assignment, 3 My released marks and feedback, 0 Log out.

To see your assignments

Step 1. Choose 1. The list shows the number of the assignment (Id), the course, your own deadline, the maximum mark and the status.

To submit your work

Put your file in a folder you can find (for example your home folder). It may be at most 5 MB.

Step 1. Choose 2. The list of your assignments shows.

Step 2. Type the **Id** of the assignment and press Enter.

Step 3. Type the **path of your file**, for example /home/student/essay.pdf, and press Enter. The path is typed here, after the question "Path of the file", and not at "Choose:".

Step 4. Read the screen: the file name, your deadline and the time now. If the work is **late**, a warning shows the days late and the penalty. Late work is accepted, but it loses marks (Table 4.1).

Step 5. Type **y** to save. A **RECEIPT** shows with the version and the time. Keep a note of it.

You can replace a file before your deadline: submit again for the same assignment. The old file is kept. After the deadline, or after the work is marked, it cannot be replaced.

Table 4.1: Late work

How late	What happens
On time or before the deadline	No penalty
1 day late (part of a day counts as a whole day)	10% of the maximum mark is taken off
2 days late	20% of the maximum mark is taken off
3 days late	30% of the maximum mark is taken off
More than 3 days late	The final mark is 0

To see your marks

Step 1. Choose 3. Only marks that were **released** are shown, with the days late, the penalty and the feedback. A mark that is not released yet is not shown.

4.3 Lecturer

Menu: 1 My courses, 2 Create an assignment, 3 My assignments, 4 Give an extension to a student, 5 See the submissions of an assignment, 6 Mark a submission, 7 Save a submitted file to this computer, 0 Log out.

To create an assignment

Step 1. Choose 2. Type the course Id, the title, the description, the deadline (2026-11-30 17:00) and the maximum mark (1 to 100).

To give an extension

Step 1. Choose 4. Type the assignment Id and the student Id (the list shows the students of the course).

Step 2. The program shows the present deadline of the student. Type the new deadline (it must be later) and the reason. The extension is for that student only. Late work of the student is worked out again with the new deadline.

To mark a submission

Step 1. Choose 6. Type the assignment Id. The submissions show with the status, the days late and the marks.

Step 2. Type the submission Id and the **raw mark** (the mark before any penalty).

Step 3. The program shows the penalty and the **final mark**. Type the feedback (it is required).

Step 4. Type **y** to save. The status is now Marked, and the Head of Department can moderate it.

To change a mark that is Marked or Returned, mark it again. The program also asks for the reason of the change. A mark that is already Moderated or Released cannot be changed by you.

To read a submitted file

Step 1. Choose 7. Type the assignment Id and the submission Id. Type the name to save the file as, or press Enter to keep its own name. Open the saved file with any program.

When the Head of Department sends a submission back, you see a message at your next log in.

4.4 Head of Department

Menu: 1 See the marked submissions of my department, 2 Approve a mark, 3 Change a final mark, 4 Send a submission back to the lecturer, 5 Release one moderated submission, 6 Release all moderated submissions of an assignment, 7 Summary of each assignment, 8 See the audit log, 9 Save a submitted file, 0 Log out.

You see only the courses of your own department. The student does not see a mark until you release it.

To moderate

Step 1. Choose 1 to read the marked work: the lecturer, the raw mark, the days late, the penalty, the final mark and the feedback.

Step 2. To accept a mark, choose 2, type the submission Id, and type **y**. The status is now Moderated.

Step 3. To change the final mark, choose 3. Type the submission Id, the new final mark and the **reason** (required). Type y. The change is written in the audit log. The status is now Moderated.

Step 4. To ask the lecturer to mark again, choose 4. Type the submission Id and a **comment** (required). Type y. The status is now Returned and the lecturer gets a message.

To release marks

Step 1. Choose 7 to see how many submissions of each assignment are Marked, Moderated and Released.

Step 2. Choose 5 to release one submission (type the assignment Id, then the submission Id, then y), or 6 to release all moderated submissions of one assignment (type the assignment Id, then y).

Only moderated work can be released. After the release the student sees the mark and the feedback, and the lecturer gets a message.

To see the audit log

Step 1. Choose 8. Every change of a final mark of your department is shown with the date, the person, the old mark, the new mark and the reason. The log cannot be changed.

4.5 Administrator

Menu: 1 List users, 2 Create a user, 3 Edit a user, 4 Deactivate a user, 5 Set a new password, 6 List departments, 7 Create a department, 8 Edit a department, 9 Deactivate a department, 10 List courses, 11 Create a course, 12 Edit a course, 13 Deactivate a course, 14 Assign a lecturer to a course, 15 Enrol a student in a course, 16 Remove a student from a course, 0 Log out.

Nobody can register alone. The Administrator makes every account.

To set up a new school year (in this order)

Step 1. Choose 7 and make the departments.

Step 2. Choose 2 and make the users. Type the username, the full name, the password twice and the role (1 Administrator, 2 Lecturer, 3 Head of Department, 4 Student). For a Head of Department, also type the department Id. Each department has one Head of Department.

Step 3. Choose 11 and make the courses (department Id, course code, title).

Step 4. Choose 14 to assign each lecturer to a course, and 15 to enrol each student.

Other tasks

- **New password for a user:** choose 5, type the user Id and the new password twice. The password needs 8 characters, a letter and a digit. Tell the user the password and ask them to keep it safe.
- **Leaving staff or students:** choose 4 to deactivate the user. Nothing is deleted, so the marks stay. You cannot deactivate your own account.
- **Deactivate a department:** only when it has no active course and no active Head of Department.
- **Backup:** see the installation guide, Section 3.5. Make a copy of lms.db each day.

5. Release notes

Moderated LMS, version 1.0. Date: 1 November 2026. The same text is in docs/RELEASE_NOTES.md in the repository.

5.1 What is in this version

- **Student:** sees the assignments with the own deadline and status; submits and resubmits a file of up to 5 MB, with a warning and the penalty for late work and a receipt; sees released marks and feedback.
- **Lecturer:** creates assignments; gives an extension to one student; marks a submission with the raw mark, the automatic late penalty (10% of the maximum mark for each day, 0 after 3 days) and the feedback; changes a mark with a reason; saves a submitted file.
- **Head of Department:** approves, changes or returns marks of the own department; releases marks one at a time or all at once; sees a summary and the audit log.
- **Administrator:** manages users, departments and courses, assigns lecturers and enrols students.
- **Security:** passwords are stored only as salted Argon2id hashes; the account is locked for 5 minutes after 5 wrong passwords; every function checks the role first; all data is changed in transactions.
- **Offline:** one program and one data file on one computer; no internet is used.

5.2 How it was checked

36 unit and integration tests, 95 system tests and 7 acceptance tests pass. Line coverage of the business layer is 85.7%. A scripted session and 60 sessions of wrong input run without problems. Static analysis (clang-tidy, cppcheck, clang-format) finds nothing. The details are in the Phase 5 and Phase 6 documents. Six defects were found and fixed during testing; none is open.

5.3 Known limits

- Built and tested on Ubuntu Linux 24.04 only. It was not tested on Windows.

- The times (login about 120 ms, saving a 5 MB file about 21 ms) were measured on a test computer, not on the reference computer.
- The trial with five people who are not IT specialists (NFR-07) and the timed installation (NFR-11) have not been done yet.
- The GitHub Actions pipeline has not run on GitHub yet; the same steps pass on the test computer.
- One user at a time; the data file is not shared between computers.
- A user cannot change the own password; the Administrator sets a new one. There is no way to recover the password of the only Administrator, so make two.
- An extension is given to one student at a time.
- Users, departments and courses can be deactivated but not deleted.

5.4 Differences from the design

The code differs from the Software Design Document in 18 small points (for example extra helper classes and dependencies). They are listed with their reasons in docs/DESIGN_FIDELITY.md and still have to be written into the design document.

5.5 Data file

The data file has the database version number 1. A file with another number is refused and not changed.

6. Version 1.0 tag

The release is the annotated Git tag **v1.0** on the main branch. The work of this phase (the release notes) was done on the branch feature/phase7-release and merged into main with git merge --no-ff before the tag was made, as in the branching strategy. The tags of the earlier phases are v0.4.0, v0.5.0 and v0.6.0. The next release is v1.1.

To get exactly this version: git clone <address> lms, then git checkout v1.0.